

# Supporting Information for Extra Hidden States under Emission Misspecification in Hidden Markov Models

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## S1 Proofs

This section gives the proofs of Lemma 2.3, Propositions 2.4–2.5, Theorem 2.6, and Corollary 2.7 from Section 2 of the main article, in that order. Notation is identical to that article. For convenience, the Kullback–Leibler divergence and  $L^1(\lambda)$  distance used below are  $D(p\|q) = \int p \log(p/q) d\lambda$  and  $\|p - q\|_1 = \int |p - q| d\lambda$ , with the extended-value conventions stated in the main article. Also let  $\Delta_K = \{(\omega_1, \dots, \omega_K) \in [0, 1]^K : \sum_{j=1}^K \omega_j = 1\}$  denote the  $K$ -weight probability simplex.

### S1.1 Proof of Lemma 2.3: separation from lower-order mixtures

The class  $\mathcal{M}_{K_0}$  is the continuous image of the compact set  $\Delta_{K_0} \times \Theta^{K_0}$  and is therefore compact in  $L^1$ . The constrained simplex defining  $\mathcal{A}_c$  is compact, so  $\mathcal{A}_c$  is compact as well. The sets are disjoint. Otherwise, a mixture containing exactly  $R$  positive-weight, pairwise distinct atoms would equal a mixture with at most  $K_0$  atoms after zero-weight and repeated atoms were removed. This contradicts identifiable minimal order. Two disjoint compact sets in a metric space have positive distance, hence

$$d_c = \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0.$$

Pinsker’s inequality,  $D(p\|q) \geq \|p - q\|_1^2 / 2$ , then gives the claimed Kullback–Leibler lower bound.

### S1.2 Proof of Proposition 2.4: aligned improvement by state splitting

In the aligned comparison, the hidden aggregate trajectory law is held fixed. All transition terms therefore cancel in the complete-data likelihood ratio. With one fitted emission  $f_{\eta_k}$  per state, stationarity gives the risk

$$\mathcal{R}(f_{\eta_1}, \dots, f_{\eta_{K_0}}) = \sum_{k=1}^{K_0} \pi_k D(g_k \| f_{\eta_k}).$$

Taking the infimum over the emission parameters gives  $\sum_k \pi_k \delta_k(1)$ . With  $m_k$  substates, the fitted aggregate emission can be any  $q_k \in \mathcal{M}_{m_k}$  by drawing a component independently at each visit

to state  $k$ , as in the refinement construction, yielding  $\sum_k \pi_k \delta_k(m_k)$ . Since  $\mathcal{M}_1 \subseteq \mathcal{M}_{m_k}$ , all terms weakly decrease. A strict decrease in one state with  $\pi_k > 0$  proves the strict inequality when the original weighted risk is finite. Without that condition another state can have infinite loss under both models, leaving both total risks infinite; only the non-increase is then guaranteed.

### S1.3 Proof of Proposition 2.5: exact HMM refinement

Let  $X_t^{\text{ref}}$  denote the refined chain and let  $r(k, a) = k$  map a refined state to its aggregate state. The refined transition matrix is stochastic because

$$\sum_{\ell, b} \gamma_{k\ell}^0 \omega_{\ell, b}^0 = 1.$$

The distribution  $\pi_k \omega_{k, a}^0$  is stationary since

$$\sum_{k, a} \pi_k \omega_{k, a}^0 \gamma_{k\ell}^0 \omega_{\ell, b}^0 = \omega_{\ell, b}^0 \sum_k \pi_k \gamma_{k\ell}^0 = \pi_\ell \omega_{\ell, b}^0.$$

Furthermore,

$$\mathbb{P}\{r(X_t^{\text{ref}}) = \ell \mid X_{t-1}^{\text{ref}} = (k, a)\} = \sum_b \gamma_{k\ell}^0 \omega_{\ell, b}^0 = \gamma_{k\ell}^0.$$

Summing the refined joint density over substate paths compatible with a fixed aggregate path  $z_{1:T} = (z_1, \dots, z_T)$  gives

$$\begin{aligned} \pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T \sum_{a=1}^{m_{z_t}} \omega_{z_t, a}^0 f_{\eta_{z_t, a}^0}(y_t) \\ = \pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T g_{z_t}(y_t). \end{aligned}$$

Summing over aggregate paths proves equality of all observed finite-dimensional distributions.

### S1.4 Proof of Theorem 2.6: observed-law separation

Exact refinement establishes

$$\inf_{Q \in \mathbf{H}_R(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Write  $\mathbb{P}_0$  and  $\mathbb{E}_0$  for probability and expectation under the true aggregate HMM. Fix a candidate  $Q \in \mathbf{H}_{K_0}(\mathcal{F})$  with emissions  $f_{\eta_1}, \dots, f_{\eta_{K_0}}$ . Let  $\beta_{t,j}^Q(y_{1:t-1})$  be its predictive probability of hidden state  $j$  given the realized observation history  $y_{1:t-1} = (y_1, \dots, y_{t-1})$ . Its one-step predictive density has the form

$$p_t^Q(y \mid y_{1:t-1}) = \sum_{j=1}^{K_0} \beta_{t,j}^Q(y_{1:t-1}) f_{\eta_j}(y),$$

where the  $\beta_{t,j}^Q$  are nonnegative and sum to one. Thus the predictive density belongs to  $\mathcal{M}_{K_0}$  for

every history, regardless of the candidate transition matrix and initial distribution.

For the true aggregate HMM, let

$$\beta_{t,k}^0(y_{1:t-1}) = \mathbb{P}_0(Z_t = k \mid Y_{1:t-1} = y_{1:t-1}).$$

Stationarity implies  $\pi_\ell = \sum_k \pi_k \gamma_{k\ell}^0 \geq \varepsilon$ , so  $\beta_{1,\ell}^0 \geq \varepsilon$ . For  $t \geq 2$ ,

$$\begin{aligned} \beta_{t,\ell}^0(y_{1:t-1}) &= \sum_k \mathbb{P}_0(Z_{t-1} = k \mid Y_{1:t-1} = y_{1:t-1}) \gamma_{k\ell}^0 \\ &\geq \varepsilon. \end{aligned}$$

The true predictive density is consequently

$$p_t^0(y \mid Y_{1:t-1}) = \sum_{k,a} \beta_{t,k}^0(Y_{1:t-1}) \omega_{k,a}^0 f_{\eta_{k,a}^0}(y),$$

with every refined weight at least  $c = \varepsilon \omega_{\min}$ . Thus  $p_t^0(\cdot \mid Y_{1:t-1}) \in \mathcal{A}_c$  almost surely. By the mixture-separation lemma,

$$D\{p_t^0(\cdot \mid Y_{1:t-1}) \parallel p_t^Q(\cdot \mid Y_{1:t-1})\} \geq d_c^2/2$$

almost surely. If a conditional divergence is infinite, the desired bound is immediate. Otherwise, the chain rule gives

$$\begin{aligned} D\{(P_0^Y)^{1:T} \parallel Q^{1:T}\} &= \sum_{t=1}^T \mathbb{E}_0 \left[ D\{p_t^0(\cdot \mid Y_{1:t-1}) \parallel p_t^Q(\cdot \mid Y_{1:t-1})\} \right] \\ &\geq T d_c^2/2. \end{aligned}$$

This proves the finite-sequence assertion for every  $T \geq 1$ . Dividing by  $T$  and taking a lower limit proves the positive relative-entropy-rate bound.

### S1.5 Proof of Corollary 2.7: enriched-emission representation

Each true emission  $g_k$  belongs to  $\mathcal{M}_M$  when  $M \geq \max_{1 \leq k \leq K_0} m_k$ . The original aggregate chain is therefore an element of  $\mathbf{H}_{K_0}(\mathcal{M}_M)$  with observed law  $P_0^Y$ , and the infimum of the relative-entropy rate is zero.

## S2 Simulation computation and diagnostics

The accompanying reproducibility files contain the simulation, genomic and supplementary movement analyses. The reproducibility repository is <https://github.com/AurelienNicosiaULaval/corrective-hmm-states>. The replicated study is regenerated from the package root with `numerics/scripts/run_optimization_audit.R`, using master seed 20260610. The compiled EM

implementation is checked against the public R updates for four component specifications, both single and multiple sequences, and three iteration counts.

The study compares Gaussian HMMs with two and three states and a two-state HMM with two Gaussian components per state. There are 200 fixed-seed replications at each of  $T \in \{500, 1500, 5000\}$ . Ten random starts use the same initial distributions as the original protocol, with 25 screening iterations for Gaussian models and 30 for mixtures, followed by refinement of the best three. Additional starts are constructed from fitted data: splitting a G2 emission or reducing an M22 emission supplies a candidate M21 model; exact M21 refinement supplies a G3 start, and duplicating its single component supplies an M22 start. The generating parameters are never used as initial values. Refinement uses relative log-likelihood tolerance  $10^{-9}$  in blocks of up to 3,000 iterations, continued until convergence. The standard-deviation floor is 0.05.

The final G3 and M22 likelihoods must be at least the fitted M21 likelihood, and all three reported fits must improve or retain their original likelihoods. The per-replication parameters, stopping diagnostics, original results and comparisons are retained in `numerics/output/optimization_audit`. These checks address observed optimization failures without proving global optimality. The R and compiled implementations add  $10^{-10}$  to initial-state and transition counts, and  $10^{-12}$  to active component counts and emission update denominators. These numerical safeguards do not change the stated model classes.

Model comparisons use the Bayesian information criterion (BIC), as defined in the main article. Row-level results are stored in `numerics/output/review_replications.csv`; the compact summary is in `numerics/output/review_summary_by_T.csv`. Additional diagnostics appear in Figures [S1–S3](#).

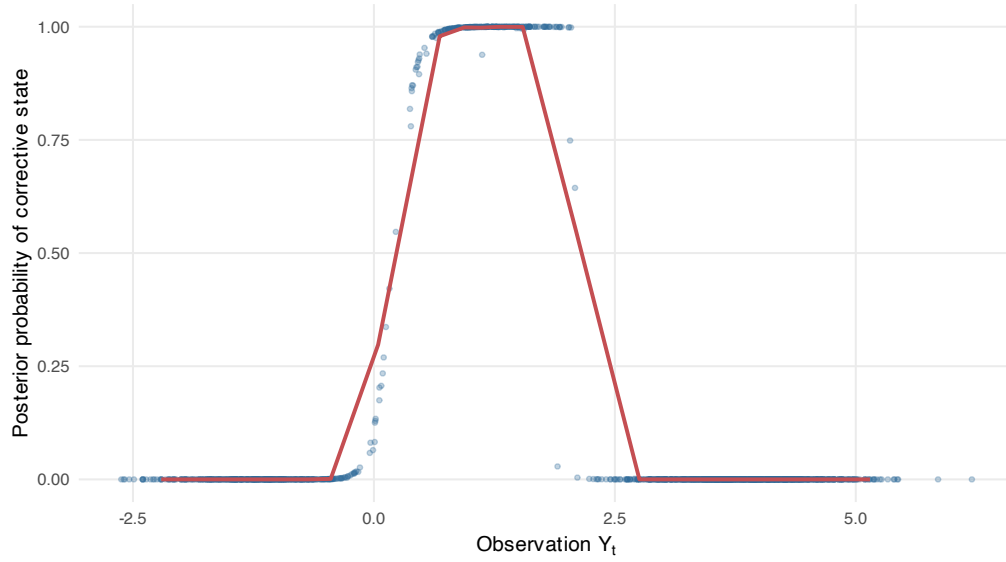


Figure S1: Posterior probability of the corrective Gaussian-HMM state in the reference  $K = 3$  fit. The probability tracks observations associated with the second component of the heterogeneous first aggregate regime.

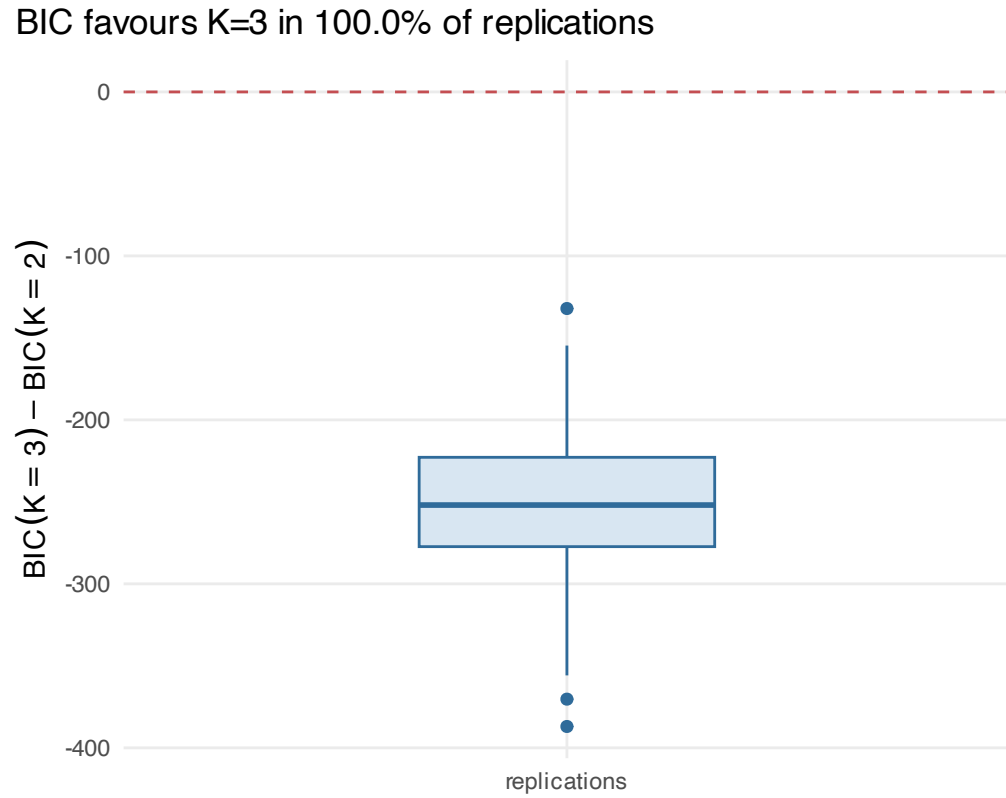


Figure S2: Distribution of the BIC difference between the three-state and two-state Gaussian models over the  $T = 1500$  replications. Negative values favour the corrective three-state Gaussian HMM.

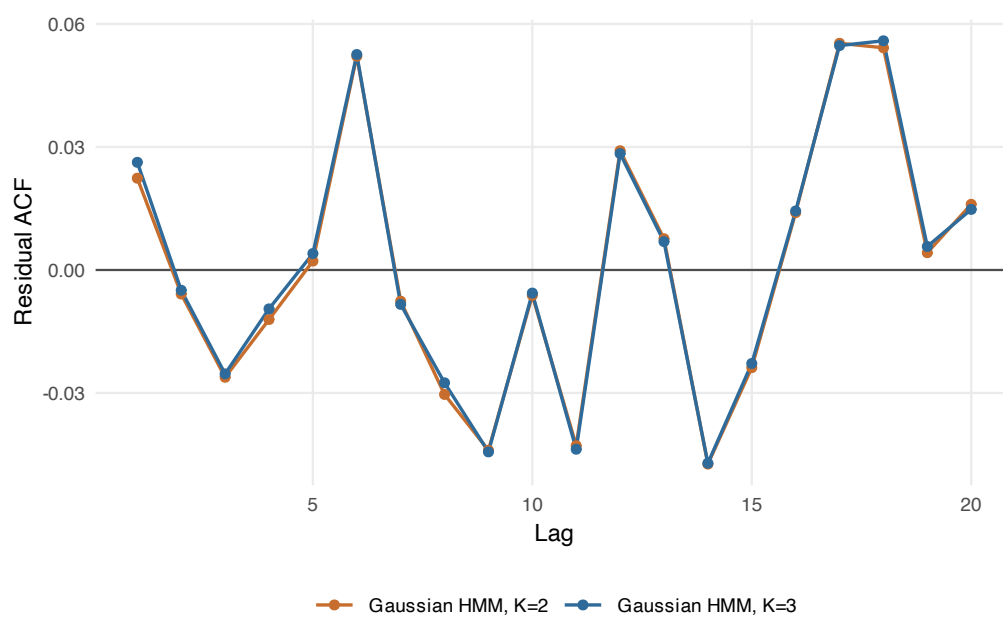


Figure S3: Autocorrelation of approximate one-step standardized prediction residuals for the Gaussian-emission HMMs with  $K = 2$  and  $K = 3$ .

## S3 Genomic application details

### S3.1 Provenance, preprocessing and partition

The original Affymetrix SNP 6.0 CEL files are from Gene Expression Omnibus (GEO), series [GSE13372](https://doi.org/10.1038/nmeth.1276), associated with Chiang et al. (2009), <https://doi.org/10.1038/nmeth.1276>. The first pair comprises GSM337641 (HCC1143) and GSM337662 (HCC1143BL); the second comprises GSM337642 and GSM337663. Their selection follows accession order, rather than a comparison of fitted profiles. Their download locations and SHA-256 hashes are recorded in the source manifest. These arrays are repeated measurements of the same cell line and matched normal, not four independent biological specimens.

AS-CRMAv2 in `aroma.affymetrix` 3.2.3 processes each array; the exported total signals and B fractions contain 1,881,415 assay units. TumorBoost v4 in `aroma.light` 3.38.0 corrects tumor fractions using the matched normal, with heterozygous normal target  $\mu_N = 0.5$  and `preserveScale=FALSE`. The procedure and correction are described by Bengtsson et al. (2009, 2010), <https://doi.org/10.1093/bioinformatics/btp371> and <https://doi.org/10.1186/1471-2105-11-245>. The matched normal serves two roles: it identifies candidate heterozygous markers, at which both A and B variants are present, and supplies the reference for marker-specific fraction correction. One observation in the HMM is the resulting corrected tumor fraction at a retained genomic position. Corrected values outside  $[0, 1]$  are possible and are retained. Gaussian measurement densities are fitted to these corrected fractions, not to bounded counts or latent allele probabilities.

Analysis coordinates come from `GenomeWideSNP_6,Full,na30,hg18,HB20100215.ugp`. The hg19 annotation used by the preprocessing environment for other purposes is not used to align the published sequencing boundaries. We require finite fractions, positive positions and matched-normal fractions in  $[0.3, 0.7]$ . At duplicate chromosome/position pairs, the first assay-unit index is retained. Chromosome 1 initially has 146,401 assay units, of which 73,346 have finite paired fraction measurements and 20,786 pass the normal-fraction filter; 20,697 unique-position markers remain. The two sequences contain 10,322 and 10,375 markers. Consecutive retained positions more than one megabase apart start a new sequence, as does every chromosome change.

The local protocol was recorded before inspecting processed profiles or fitting HMMs. Chromosome 1 was designated for initial development; chromosomes 2–11 were reserved as possible extensions but were unused. Chromosomes 12–17 were validation, and 18–22 were not evaluated until the fitted model files and specification had been hashed and frozen. This is a local exploratory design, not an externally registered confirmatory study. The saved model objects and original protocol match the recorded freeze hashes. The original protocol is preserved in `genomics/audit/protocol_before_confirmation.md`; the current protocol guide has been edited for presentation. The distributed evaluation script is a later version. Here “confirmation” denotes a later held-out block within this exploratory analysis, rather than preregistered confirmatory inference. No chromosome outside chromosome 1 contributes to parameter estimation. The held-out blocks contain 52,294 markers on chromosomes 12–17 and 25,413 on chromosomes 18–22. The

second array pair supplies 20,937 chromosome-1 markers, including 19,727 positions shared with the first pair, and assesses technical repeatability.

The seven original candidates were G2, G3, G5, M21, M221, F2 and F3. G4 was added on 24 September 2026 in response to the reviewer, after the original evaluation results had been inspected. It uses the same chromosome-1 input and is evaluated on the same fixed blocks. Its results are therefore a retrospective extension of the comparison, not part of the original model freeze or a new confirmatory analysis. The original fitted objects and freeze record are retained without alteration.

### S3.2 Estimation and numerical checks

All ordinary and unconstrained mixture models use the EM updates in the main article. The likelihood is a product over explicitly reset sequences; initial-state updates average responsibilities at all sequence starts, and transition updates exclude sequence boundaries. The standard-deviation floor is 0.01 on the corrected-fraction scale. The ordinary and unconstrained mixture implementations use the numerical pseudocounts stated in SI S2. Forty random starts receive up to 100 screening iterations at relative tolerance  $10^{-5}$ , followed by up to 2,500 iterations at  $10^{-8}$  for the best eight. Persistent initializations use diagonal transition probabilities 0.95, 0.995 and 0.9995. For G2 these include a common centre of 0.5 with different variances: allowing only separated-mean starts missed the higher-likelihood variance-regime fit. Mixture starts supplement random initializations with low/high pairs. Selections are always based on likelihood, not on appealing decoded labels.

G2 is included in M21 by duplicating an emission component. M21 is included in G3 by the exact refinement; G3 is included in M221 by duplicating the first two component distributions. A duplicated aggregate state additionally embeds M21 in M221. Finally, M221 is included in G5 by exact refinement. The code checks the likelihood at these inclusions and refines from them, including small separation perturbations. The selected G3 and G5 likelihoods must be at least those of M21 and M221. G4 uses the same 40-start screening and eight-start refinement settings, with seed 20261314. Each G3 state is also duplicated in turn to create an exactly equivalent G4 start. The two duplicated means are then either kept equal or perturbed by  $\pm 0.015$ , giving nine additional starts. All nine are refined, rather than discarded at the screening stage. Its selected log likelihood lies between those of G3 and G5. The implementation uses the compiled equivalent of the R EM updates; the selected likelihood is checked independently with the R emission calculation and sequence recursion. Parameters, complete start-level likelihood traces and input hashes are retained in `genomics/results/genomic_g4`. The tests independently compare the multiple-sequence recursion with direct path enumeration and verify that changing later observations cannot alter earlier predictive probabilities.

The unrestricted five-state model has a minimum transition estimate of about  $1.29 \times 10^{-7}$ . This is not a zero transition or proof of structural absence. G4 has a smaller minimum transition estimate,  $2.60 \times 10^{-14}$ , effectively at the boundary at the reported precision. We retain this diagnostic rather than treating its BIC as a regular-model order test. The lower variance bound is inactive in every selected primary fit. Relative convergence of EM and multiple starts do not prove global optimality;

Table S1: Selected estimates on chromosome 1. Runs count consecutive Viterbi labels within each sequence; they are not verified genomic alterations. Convergence refers to the final relative likelihood stopping criterion, not to proof of a global optimum.

| Model | $\ell$    | Runs | Converged | Floor hits |
|-------|-----------|------|-----------|------------|
| G2    | 7410.531  | 14   | Yes       | 0          |
| G3    | 10438.681 | 4009 | Yes       | 0          |
| G4    | 11321.846 | 3497 | Yes       | 0          |
| G5    | 12517.423 | 9123 | Yes       | 0          |
| M21   | 10436.225 | 78   | Yes       | 0          |
| M221  | 12508.295 | 181  | Yes       | 0          |
| F2    | 10980.247 | 68   | Yes       | 0          |
| F3    | 12487.146 | 184  | Yes       | 0          |

all screened and refined likelihoods are retained in the fitted objects.

**Likelihood-ratio calibration for the refinement constraints.** The prespecified grouping has two paired emissions and one single emission. Holzmman and Schwaiger (2015), <https://doi.org/10.1007/s11222-014-9481-1>, give a state-reduction test under a stationary initial law. Their reference degrees of freedom for reducing  $k$  states to  $r$  groups are  $k^2 - 2k - r^2 + 2r$ , hence 12 for  $k = 5$  and  $r = 3$ . Here the common initial law is freely estimated from two sequences. Increasing their lengths does not increase the number of initial draws. Adding the two extra initial-law parameters therefore does not justify replacing 12 by 14 chi-squared degrees of freedom. The likelihood ratio compares M221 with its unrestricted five-state Gaussian refinement, including the corresponding initial-distribution constraints. The statistic after numerical refinement is 18.2582; the saved primary fits give 18.2572, a difference of approximately 0.001. The refined null and alternative log likelihoods improve by less than 0.001 each and leave the reported comparison unchanged at its stated precision.

The parametric bootstrap uses 199 independent simulations from the refined M221 fit, with sequence lengths 10,322 and 10,375. Each sequence starts from the common fitted initial distribution. Simulation seeds are  $202609170 + b$ ,  $b = 1, \dots, 199$ , with fitting seeds offset by 10,000. For both observed and simulated data, each model screens ten random starts and starts at the saved primary parameters for 100 iterations. These saved parameters come from the observed data: they provide preoptimized starts for the observed series but serve as warm starts for each simulated series. The best four are refined at relative likelihood tolerance  $10^{-8}$ ; the alternative also retains and refines the exact expansion of its fitted null. All selected fits satisfy the stopping criterion and all alternative likelihoods are at least their null likelihoods. Parameters, traces and seeds are saved in `genomics/results/refinement_bootstrap`; the executable script is `genomics/scripts/genomic_refinement_bootstrap.R`.

Seventeen simulated statistics meet or exceed the observed value. The plus-one estimate is 0.09, with Monte Carlo standard error 0.020; the exact 95% binomial interval for the underlying bootstrap tail probability is [0.051, 0.133]. This interval describes Monte Carlo uncertainty only.

For comparison, the bootstrap mean and 95% quantile are 12.3 and 21.1; the corresponding  $\chi^2_{12}$  values are 12 and 21.0, and its tail probability at 18.2582 is 0.108. Agreement of these summaries is descriptive; it does not establish the assumptions of the asymptotic reference law. The bootstrap is conditional on M221 being an adequate observation model; it does not correct residual dependence or validate a biological partition. Because the initial distribution is estimated from two sequence starts, we do not use the difference of 14 free parameters as automatic chi-squared degrees of freedom. The stationary-law result concerns different initial-law assumptions. We report the bootstrap as an approximate calibration of this specified constrained comparison, not as a new state-reduction test.

### S3.3 Prediction and calibration

Sequential held-out log scores use the previous forward filter and fixed training parameters. Each validation or confirmation sequence starts with the fitted initial distribution, not a state estimated from the full held-out path. Smoothed states are used only for descriptive assignment and are never substituted into the predictive density. Table S3 gives chromosome-specific differences, including those favouring G4 and F3. Table S2 preserves the full eight-model comparison.

Table S2: Full genomic comparison. All parameters are estimated on chromosome 1. Lower BIC and higher sequential log scores are preferred in their respective columns. F2/F3 impose reflection symmetry. G4 was added retrospectively after the original seven-model evaluation; it was not part of the original model freeze.

| Model | $p$ | BIC      | Repeat   | Chr. 12–17 | Chr. 18–22 |
|-------|-----|----------|----------|------------|------------|
| G2    | 7   | -14751.5 | 7049.59  | -46.97     | 15572.46   |
| G3    | 14  | -20738.2 | 9818.62  | 14410.44   | 17013.16   |
| G4    | 23  | -22415.1 | 10767.74 | 17978.14   | 17371.75   |
| G5    | 34  | -24697.0 | 11907.57 | 16350.04   | 21365.91   |
| M21   | 10  | -20773.1 | 9816.15  | 14423.18   | 17015.30   |
| M221  | 20  | -24817.8 | 11906.57 | 16386.29   | 21371.08   |
| F2    | 7   | -21890.9 | 10309.49 | 14620.20   | 16146.25   |
| F3    | 14  | -24835.2 | 11887.64 | 16379.29   | 21386.96   |

Relative to G5, M221 gains 36.25 log-score units on chromosomes 12–17 and 5.16 on chromosomes 18–22, but loses 0.99 on the technical repeat. The corresponding differences per marker are 0.00069, 0.00020 and  $-0.00005$ . These are small differences in this comparison, without a prespecified equivalence margin or sampling uncertainty estimate. G4 has the highest score on chromosomes 12–17, exceeding M221 by 1591.85, although its chromosome-1 BIC and technical-repeat score are worse. On chromosomes 18–22, its score is 3999.33 lower than M221’s. Thus the chromosome-1 preference does not imply a uniform predictive ranking. The G4 extension is retrospective, as stated in S3.1.

The empirical PIT distance is  $\max_i \{i/n - u_{(i)}, u_{(i)} - (i-1)/n\}$ , without an independent-observation Kolmogorov–Smirnov  $p$ -value. Central predictive coverage is the fraction of PIT values in  $[0.025, 0.975]$ . Normal-score residuals use inverse-normal PITs, bounded numerically away from 0 and 1 by  $10^{-12}$ . Their absolute values retain lag-one correlations of 0.057 and 0.142 for M221

Table S3: Sequential log-score differences by excluded chromosome, using fixed chromosome-1 estimates. Differences are comparator minus M221, so positive values favour the comparator. The G4 comparison was added retrospectively.

| Chromosome | $n$   | G4 – M221 | G5 – M221 | F3 – M221 |
|------------|-------|-----------|-----------|-----------|
| 12         | 12637 | 974.19    | -10.08    | -10.33    |
| 13         | 9646  | 298.64    | -1.51     | -4.71     |
| 14         | 8395  | 981.36    | -3.74     | 0.04      |
| 15         | 7493  | -610.50   | -12.15    | 1.26      |
| 16         | 8071  | -767.94   | -6.12     | 6.18      |
| 17         | 6052  | 716.10    | -2.64     | 0.55      |
| 18         | 7766  | -621.00   | -1.19     | 4.06      |
| 19         | 3823  | -760.30   | 0.78      | -0.21     |
| 20         | 6482  | -1870.22  | 5.83      | 14.97     |
| 21         | 3767  | -30.02    | -10.93    | -1.52     |
| 22         | 3575  | -717.79   | 0.36      | -1.41     |

Table S4: Fixed-parameter predictive diagnostics. Coverage is the percentage inside the central 95% predictive interval;  $D_{\text{PIT}}$  is the empirical distance from a uniform PIT distribution. The last two columns are lag-one correlations of normal-score residuals and their absolute values, excluding sequence boundaries.

| Chromosomes | Model | Coverage (%) | $D_{\text{PIT}}$ | $r_1$  | $r_1^{ \cdot }$ |
|-------------|-------|--------------|------------------|--------|-----------------|
| 12–17       | G2    | 98.92        | 0.131            | -0.000 | 0.249           |
| 12–17       | G3    | 97.85        | 0.062            | -0.005 | 0.132           |
| 12–17       | G4    | 97.03        | 0.036            | -0.008 | 0.052           |
| 12–17       | G5    | 96.28        | 0.055            | -0.006 | 0.063           |
| 12–17       | M21   | 97.83        | 0.062            | -0.000 | 0.130           |
| 12–17       | M221  | 96.36        | 0.055            | -0.001 | 0.057           |
| 12–17       | F2    | 97.06        | 0.054            | 0.001  | 0.139           |
| 12–17       | F3    | 96.30        | 0.062            | -0.001 | 0.056           |
| 18–22       | G2    | 98.36        | 0.034            | -0.007 | 0.454           |
| 18–22       | G3    | 98.10        | 0.037            | -0.009 | 0.397           |
| 18–22       | G4    | 98.08        | 0.034            | -0.012 | 0.355           |
| 18–22       | G5    | 95.40        | 0.011            | -0.012 | 0.148           |
| 18–22       | M21   | 98.11        | 0.036            | -0.006 | 0.396           |
| 18–22       | M221  | 95.48        | 0.012            | -0.006 | 0.142           |
| 18–22       | F2    | 96.55        | 0.066            | -0.005 | 0.383           |
| 18–22       | F3    | 95.52        | 0.007            | -0.007 | 0.141           |

on validation and confirmation chromosomes. Thus small signed correlations and near-nominal coverage do not imply conditional independence or complete calibration.

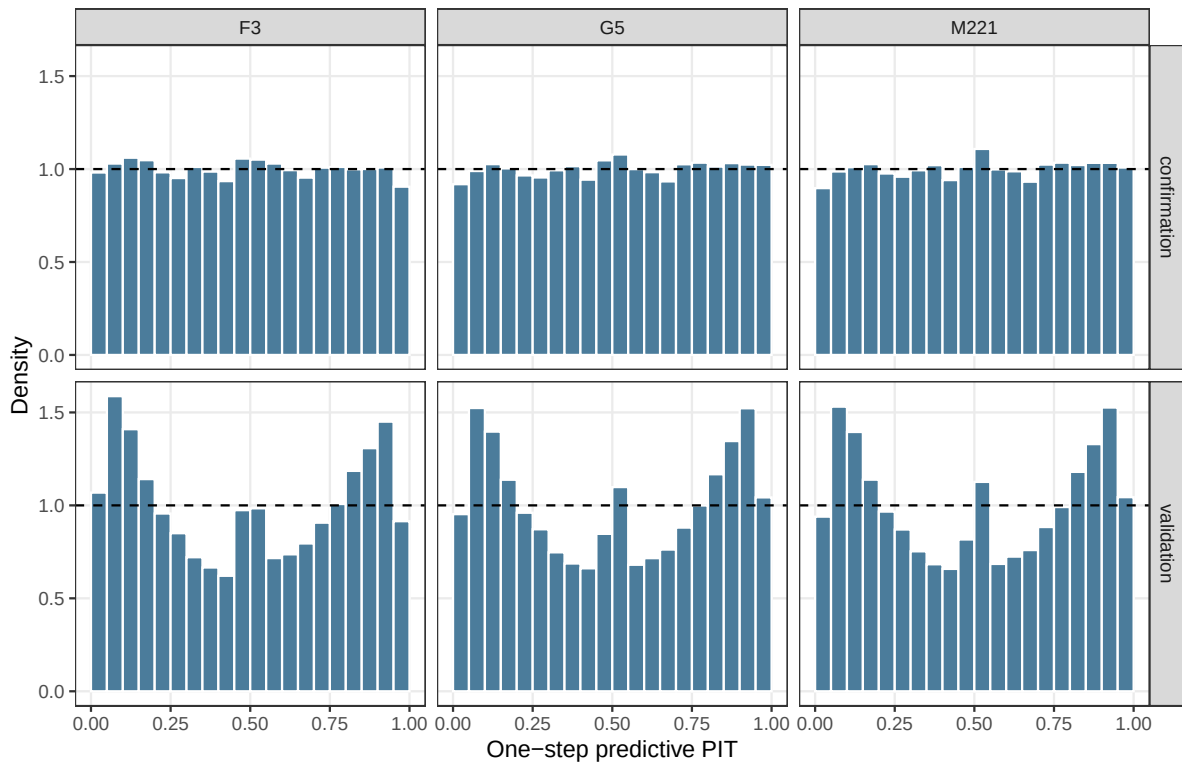


Figure S4: Predictive PIT values from fixed chromosome-1 estimates. Each row uses chromosomes excluded from fitting. The dashed line is the uniform density. The figure shows remaining departures as well as similarities between the mixture, ordinary and symmetry-aware models.

### S3.4 Aggregation, technical replication and sensitivity

Correspondence between the five G5 states and M221 components minimizes the sum of squared Hellinger distances between Gaussian emissions. The assignment uses chromosome-1 emission estimates only; the same map is applied to the technical replicate and held-out chromosomes. The largest matched distance is below  $10^{-5}$ . Aggregated G5 and M221 have 99.73% agreement on the estimation markers, 99.07% on validation and 99.06% on confirmation. The unaggregated G5 labels are not interpreted as five distinct regional levels of allelic imbalance.

Technical replication compares fixed-parameter Viterbi labels at the 19,727 positions shared by the two separately filtered chromosome-1 arrays. M221 agreement is 98.08%; F3 agreement is 98.04%. These values measure repeatability of the fitted assignments on these arrays. They do not estimate accuracy against independently known states.

Sensitivity fits use ten random starts, refinement of the best three for up to 2,000 iterations, and the primary solution as an additional start. For thinning by a factor  $q$ , the warm-start transition matrix is  $\Gamma^q$ ; parameters are then reestimated. Gaps are evaluated on the retained marker positions,

Table S5: M221 sensitivity. Agreement is evaluated at markers shared with the primary analysis; parameters are refitted within each scenario.

| Scenario                     | $n$   | Agreement (%) | Runs |
|------------------------------|-------|---------------|------|
| $\sigma_{\min} = 0.005$      | 20697 | 100.00        | 181  |
| $\sigma_{\min} = 0.020$      | 20697 | 100.00        | 181  |
| Every second marker          | 10349 | 97.56         | 33   |
| Every fifth marker           | 4140  | 96.35         | 20   |
| Normal fraction [0.25, 0.75] | 23611 | 98.58         | 302  |
| Normal fraction [0.35, 0.65] | 18558 | 98.76         | 55   |

with the same one-megabase reset rule. Agreement is computed only at markers present in both versions. Filters can change which marker is kept at a duplicated position, explaining why the common counts are sometimes slightly below the smaller sample size. All selected sensitivity fits converge and have no variance-floor hits. Changing the standard-deviation floor to 0.005 or 0.02 does not change assignments; both values lie below the smallest primary fitted standard deviation, approximately 0.054, so these constraints are inactive. The alternative normal-fraction filters retain 98.58% and 98.76% assignment agreement at shared markers, but change the run count from 181 to 302 and 55. Retaining every second or fifth marker gives 97.56% and 96.35% agreement, with only 33 and 20 runs. Broad regional profiles are more stable than individual short segments. The 181 primary runs must not be read as 181 verified alterations, nor the transition probabilities as physical segment lengths.

The allele-designation experiment uses seed 20260910 to select each marker independently with probability one half for the transformation  $y_i \mapsto 1 - y_i$ , without refitting. The fixed M221 and G5 log likelihoods change by  $-44.23$  and  $-66.54$ , respectively. The M221 state path agrees on 99.87% of markers, the raw G5 path on 56.63%, and aggregated G5 on 99.55%. F2 and F3 are invariant up to numerical rounding. Approximate invariance of the unconstrained mixture is an empirical finding, not an imposed property.

### S3.5 Models incorporating allelic symmetry

The complementary low and high fractions motivate a separate constrained family with state densities

$$f_k(y) = \frac{1}{2}\phi(y; \frac{1}{2} - d_k, s_k^2) + \frac{1}{2}\phi(y; \frac{1}{2} + d_k, s_k^2), \quad d_k \geq 0.$$

Here  $\phi(y; \mu, \sigma^2)$  denotes the normal density. The two-state F2 and three-state F3 models are equivalent to folded-normal models for  $|y - 1/2|$ . They have four and six tied Gaussian components, with 7 and 14 free parameters. F3 is a separate comparator; it is not the equal-component or nested counterpart of M221. Table S2 reports both models alongside the unconstrained candidates.

For F2 and F3, the E-step uses the two reflected normal contributions within each state. The M-step ties their variances, weights them equally and updates a nonnegative separation from one half. Thirty starts receive 120 screening iterations; the best six are refined for up to 3,000 iterations

at tolerance  $10^{-8}$ . Initial and transition probabilities have a  $10^{-6}$  lower bound. Scores are densities of the original signed fraction, so no folded-density factor is inadvertently added to comparisons with G2–G5 or M21/M221. Direct enumeration, likelihood ascent and exact invariance under independent allele flips are tested. F3’s fitted separations are 0.0041, 0.1048 and 0.3101. The nearly zero first separation is another reason not to use regular asymptotic mixture-order tests.

F3 has BIC 17.4 below M221, although its training log likelihood is 21.15 lower. Its score on chromosomes 18–22 is 15.88 higher, with 14.97 of that difference arising on chromosome 20. Its scores on chromosomes 12–17 and on the technical repeat are 7.00 and 18.93 lower, respectively. It reproduces the same broad three-profile pattern, with 98.04% technical-repeat assignment agreement compared with 98.08% for M221. The known symmetry can therefore provide a more parsimonious description, while its predictive advantages vary across sequences. The comparison is retained as evidence against a general superiority claim for M221.

### S3.6 Genomic comparators and limits of the external reference

PSCBS 0.68 uses the same candidate heterozygous loci and the total-signal ratio  $2T/N$ , with the two chromosome-1 sequences analyzed separately. We use the default paired segmentation procedure with seeds 20260911 and 20260912. It returns 51 segments. The factor two is the method’s relative copy-number convention; it does not establish absolute tumor ploidy. PSCBS combines the total signal with allelic imbalance, so it is a comparator using additional information beyond the HMM allele fractions. At markers assigned to the approximately balanced, moderate and strong M221 profiles, its median allelic-imbalance measure is 0.080, 0.214 and 0.641, respectively. This supports the ordering of the fitted profiles, but is not independent validation because PSCBS uses the same arrays. See Olshen et al. (2011), <https://doi.org/10.1093/bioinformatics/btr329>.

The publisher’s supplementary sequencing tables for Chiang et al. (2009) contain 407 HCC1143 segments, extracted with chromosome, start, end, relative copy ratio and read counts. The original PDF and its hash are recorded; selected coordinates were checked against the rendered source. The HCC1143 chr1 interval 210,542,048–221,670,720 contains 1,152 retained SNPs. Of these, 1,032 are assigned to the approximately balanced M221 profile, 116 to moderate imbalance and four to strong imbalance. Published relative copy ratios range from 1.438 to 2.255 across the interval. In the following published segment, 221,670,740–233,147,476, the relative copy ratio is 1.072, although the allele fractions are predominantly imbalanced. Total dosage and allelic balance therefore need separate interpretation.

These sequencing measurements were obtained independently of the arrays, but they concern the same cell line and measure a different quantity. They cannot supply true labels for the three HMM profiles. The displayed interval is an illustration on the development chromosome, selected to explain that distinction; it is not presented as a reserved validation endpoint. No clinical claim, absolute copy-number estimate, patient-level performance claim or novel biological discovery follows from this comparison.

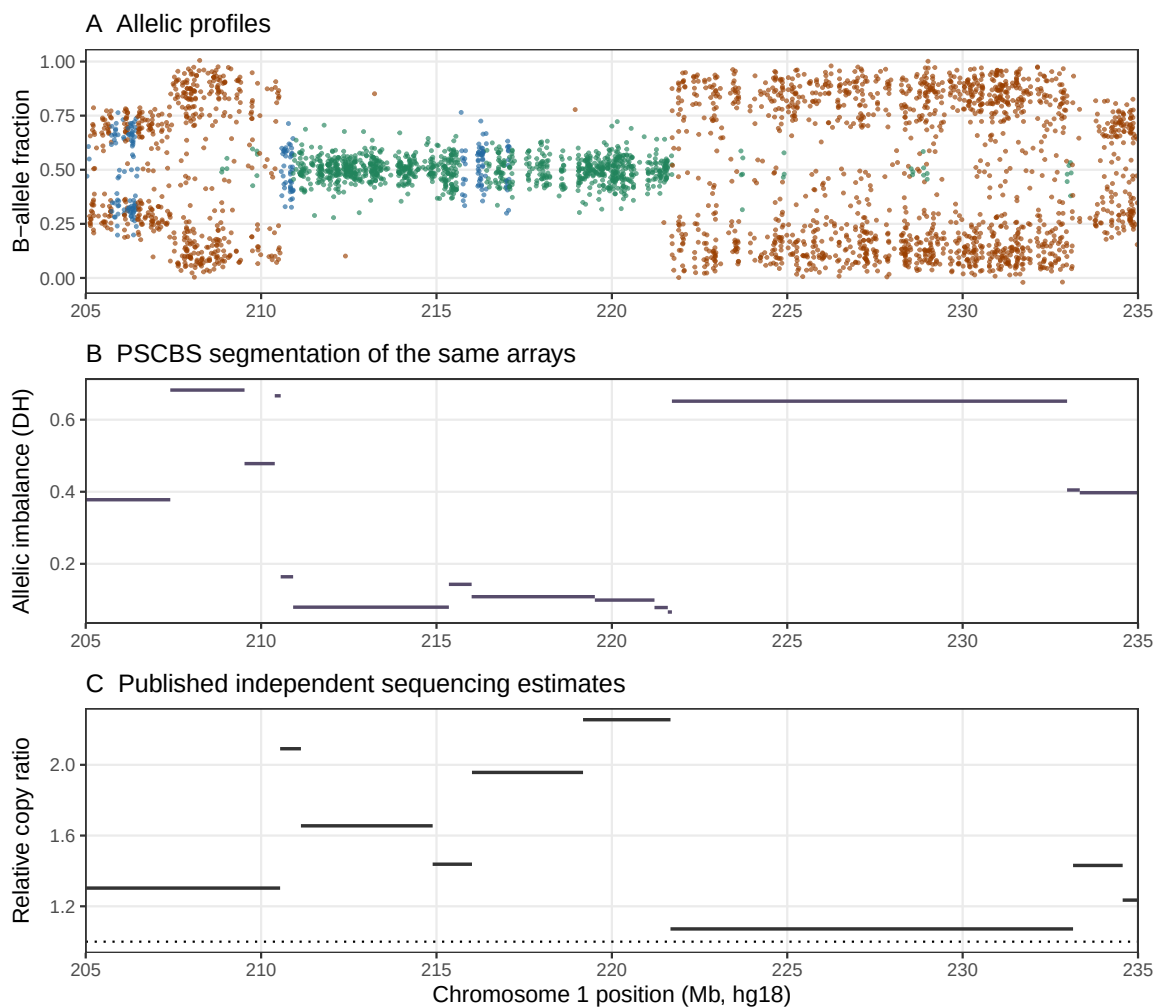


Figure S5: A chromosome-1 interval used to compare measurements. Panel A shows the M221 allele-fraction profiles. Panel B shows PSCBS segmentation of the same arrays using total intensity and allele fractions. Panel C shows the independent sequencing copy-ratio segments of Chiang et al. (2009). Approximately balanced fractions can coexist with increased total DNA dosage. This comparison does not provide true labels for the HMM states.

## S4 Elk movement: Gaussian comparisons and limitations

Elk movement provides a contrasting case: flexible Gaussian emissions can improve distributional fit without establishing a behavioral interpretation. We examine density modes, fitting constraints, temporal prediction and alternative observation families. The short series and absence of observed behavior labels limit the interpretation of the fitted states.

### S4.1 Input data and matched supplementary measurements

The data comprise GPS locations of four elk translocated in east-central Ontario, Canada, studied by Morales et al. (2004), <https://doi.org/10.1890/03-0269>. The analysis reads `elk_data` from `moveHMM` version 1.10 (Michelot et al., 2016, <https://doi.org/10.1111/2041-210X.12578>) and checks its structure and serialized SHA-256 digest before fitting. Consecutive Euclidean displacements are computed separately for each animal, converted from metres to kilometres and transformed by  $\log(1 + x)$ . The same 735 locations yield 731 within-animal displacements. Exact observation times are unavailable, so the transition step is a recorded interval, treated as approximately daily, rather than an exactly measured day.

The original study’s supplementary archive <https://doi.org/10.6084/m9.figshare.3523667.v1> provides turning angles, interval-adjusted daily movement rates and habitat measurements. All 735 animal IDs and coordinate pairs match the package data exactly. The archived text file also contains 370 empty rows, which are removed before this check. The file hash, URL and alignment checks are recorded in `empirical/revision2/context_provenance.json`. Auxiliary quantities are aligned to the departure fix of each outgoing displacement. The first turning angle of each animal is unavailable. These measurements were not fitted as responses or used to choose starting values in the Gaussian benchmark. Section S5 fits the rates and angles jointly in a separate, support-respecting analysis.

For elk 163, posterior-weighted mean turning-angle cosines in the M21 states are  $-0.232$  and  $-0.261$ , respectively. Their mean distances to open forest are 17.40 and 10.86 km. Neither comparison validates a behavioural label: angles share the trajectory’s measurement errors, while distance to habitat also changes with the animal’s geographic location and does not measure habitat selection against availability. The full summaries for every fitted state and animal are in `empirical/revision2/context_summary.csv`. The Gaussian benchmark statement that the regimes differ in displacement should therefore not be read as a demonstrated contrast in foraging, travelling or habitat preference.

A targeted check for elk 163 replaces the derived displacements with the published interval-adjusted rates for the same 158 intervals. With the same transformation, bound and fitting protocol, the BIC values are 135.1 (G2), 116.0 (G3), 139.8 (G4), 109.1 (M21) and 112.0 (M22). Thus M21 retains a 7.0-point BIC advantage over G3. This check addresses the use of approximate rather than exact time intervals; it is not an additional independent data set. Its fitted objects and numerical results are supplied.

## S4.2 Estimation, local solutions and boundary checks

G2, G3 and G4 have 7, 14 and 23 free parameters; M21 and M22 have 10 and 13. All models estimate an initial distribution and a row-stochastic transition matrix. The primary bound on component standard deviation is 0.05, with 0.025 and 0.10 used for sensitivity. Each fit starts from 200 seeded random initializations. After at most 70 screening iterations (relative likelihood tolerance  $10^{-4}$ ), the best 20 runs are refined at tolerance  $10^{-8}$ . Refinement continues in blocks of 2000 iterations, up to 10,000 if needed. Unrestricted G3 and G4 fits also start at the exact expansion of their M21 and M22 counterparts. Full-series estimates at the three bounds initialize one another for three additional passes. These warm starts use only the same full series and are not passed to training-prefix fits.

Every selected fit in the 100-model primary, sensitivity and validation grid satisfied the stopping rule. All five published-rate fits also converged. All start-level likelihoods, iteration counts and final increments are retained. The best refined solution was not always reached by many starts: for example, the final full-series M21 solution for elk 363 was found by a cross-bound warm start after the initial screening missed it. This changed its BIC from 170.8 to 157.4. Reporting convergence alone would miss this optimization sensitivity. Nested likelihood inequalities and nondecreasing likelihood as the standard-deviation bound is relaxed are checked explicitly. These checks do not certify a global optimum.

The variance-bound counts in Table S6 concern emission standard deviations. Small transition estimates are reported separately; initial probabilities may also approach the boundary because only one trajectory is fitted per animal. The near-absorbing M22 state for elk 287 is particularly consequential for interpreting switching. It is retained as a limitation, rather than suppressed by an additional transition constraint.

## S4.3 Predictive assessment

For a series of length  $T$ , the training prefixes end at  $n_1 = \lfloor 0.6T \rfloor$  and  $n_2 = \lfloor 0.8T \rfloor$ . The first fit scores  $n_1 + 1, \dots, n_2$ , and the second scores  $n_2 + 1, \dots, T$ . Filtering in each block uses the fitted initial distribution and all preceding observations, with parameters fixed at their training estimates. No full-series parameter estimates are used to initialize or score a training-prefix fit. The blocks are disjoint as scored observations, although the second training set contains the first validation block. They are consequently not independent replicates for a significance test.

The log-score advantage for elk 163 is not uniform across blocks. M21 and G3 score  $-7.751$  and  $-7.603$  on the first 32 reserved observations, and  $7.775$  and  $4.408$  on the final 32. M22 scores  $-8.056$  and  $9.785$ . All per-observation forecasts, PIT values and negative-support probabilities are saved in `empirical/revision2/heldout_predictions.csv`; block totals are in `empirical/revision2/heldout_by_fold.csv`. The retained Gaussian PIT curves and the table use these reserved observations. Full-data density plots remain descriptive fits and are not predictive validation.

The transformed response is nonnegative, whereas Gaussian approximations have real-line support. Table S7 reports the mean fitted predictive mass below zero. This mass is not removed by

Table S6: Numerical audit at bound 0.05. Gain is the final relative likelihood increment; AtBound counts component standard deviations at the bound. MinTrans and MaxSelf summarize transition boundaries. NearBest counts original refined starts within 0.01 log-likelihood units of the final selected solution, including subsequent warm-start improvements. Convergence of EM does not establish a global maximum.

| Elk | Model | Gain    | AtBound | MinTrans | MaxSelf | NearBest |
|-----|-------|---------|---------|----------|---------|----------|
| 115 | M21   | 5.5e-10 | 0       | 0.0416   | 0.9584  | 1/20     |
| 115 | M22   | 2.8e-09 | 1       | 0.0999   | 0.9001  | 20/20    |
| 115 | G2    | 2.9e-10 | 0       | 0.1117   | 0.8883  | 20/20    |
| 115 | G3    | 3.7e-09 | 1       | 0.0244   | 0.8908  | 20/21    |
| 115 | G4    | 4.1e-09 | 1       | 0.0000   | 0.6965  | 20/21    |
| 163 | M21   | 6.6e-09 | 1       | 0.2130   | 0.7870  | 20/20    |
| 163 | M22   | 4.0e-09 | 2       | 0.2367   | 0.7633  | 20/20    |
| 163 | G2    | 3.2e-10 | 0       | 0.2314   | 0.7686  | 20/20    |
| 163 | G3    | 5.2e-09 | 1       | 0.1640   | 0.6688  | 21/21    |
| 163 | G4    | 3.7e-09 | 2       | 0.0000   | 0.6372  | 17/21    |
| 287 | M21   | 3.4e-09 | 0       | 0.1099   | 0.8901  | 20/20    |
| 287 | M22   | 9.1e-09 | 0       | 0.0000   | 1.0000  | 20/20    |
| 287 | G2    | 2.4e-10 | 0       | 0.1292   | 0.8708  | 20/20    |
| 287 | G3    | 2.5e-09 | 0       | 0.0000   | 1.0000  | 21/21    |
| 287 | G4    | 7.8e-09 | 0       | 0.0000   | 1.0000  | 20/21    |
| 363 | M21   | 7.1e-10 | 0       | 0.0603   | 0.9397  | 0/20     |
| 363 | M22   | 1.2e-09 | 0       | 0.0980   | 0.9020  | 4/20     |
| 363 | G2    | 4.1e-10 | 0       | 0.3104   | 0.6896  | 19/20    |
| 363 | G3    | 8.7e-10 | 0       | 0.0000   | 0.8041  | 20/21    |
| 363 | G4    | 9.0e-09 | 1       | 0.0000   | 1.0000  | 19/21    |

Table S7: Held-out predictive summaries over two disjoint blocks: observations after the first 60% through 80%, and after 80% through the end. Each block uses parameters fitted to its preceding prefix. Coverage is the fraction of PIT values in  $[0.05, 0.95]$ ; PIT distance is the empirical Kolmogorov distance.  $P(Y < 0)$  is the mean predictive probability assigned below the support of the transformed response. Higher log scores are better.

| Elk | Model | $n_{\text{test}}$ | Log score | Coverage | PIT distance | $P(Y < 0)$ |
|-----|-------|-------------------|-----------|----------|--------------|------------|
| 115 | G2    | 78                | -51.85    | 0.949    | 0.120        | 0.061      |
| 115 | G3    | 78                | -47.82    | 0.872    | 0.124        | 0.035      |
| 115 | G4    | 78                | -51.34    | 0.936    | 0.081        | 0.039      |
| 115 | M21   | 78                | -50.41    | 0.846    | 0.112        | 0.030      |
| 115 | M22   | 78                | -51.30    | 0.846    | 0.107        | 0.027      |
| 163 | G2    | 64                | -11.49    | 0.984    | 0.172        | 0.081      |
| 163 | G3    | 64                | -3.20     | 0.984    | 0.265        | 0.060      |
| 163 | G4    | 64                | -9.83     | 0.969    | 0.276        | 0.046      |
| 163 | M21   | 64                | 0.02      | 0.969    | 0.172        | 0.048      |
| 163 | M22   | 64                | 1.73      | 0.938    | 0.185        | 0.050      |
| 287 | G2    | 66                | -52.39    | 1.000    | 0.246        | 0.044      |
| 287 | G3    | 66                | -21.03    | 1.000    | 0.173        | 0.040      |
| 287 | G4    | 66                | -52.45    | 0.985    | 0.178        | 0.024      |
| 287 | M21   | 66                | -42.72    | 1.000    | 0.219        | 0.035      |
| 287 | M22   | 66                | -24.33    | 0.985    | 0.228        | 0.019      |
| 363 | G2    | 87                | -42.80    | 0.874    | 0.404        | 0.039      |
| 363 | G3    | 87                | -28.21    | 0.667    | 0.365        | 0.013      |
| 363 | G4    | 87                | -37.12    | 0.632    | 0.383        | 0.010      |
| 363 | M21   | 87                | -32.15    | 0.655    | 0.385        | 0.012      |
| 363 | M22   | 87                | -35.17    | 0.632    | 0.371        | 0.010      |

renormalizing the predictive density after fitting. Accordingly, all model scores refer to the same real-line observation model, and the support mismatch remains visible as a limitation.

#### S4.4 Components, modes and state stability

Table S8: Component parameters at bound 0.05. Means and standard deviations are on the transformed scale. Occupation is the average smoothed state probability and Self is the state self-transition probability. Components within a state share the same transition dynamics.

| Elk | Model | State | Component | Mean  | SD    | Weight | Occupation | Self  |
|-----|-------|-------|-----------|-------|-------|--------|------------|-------|
| 115 | M21   | 1     | 1         | 0.458 | 0.243 | 1.000  | 0.517      | 0.958 |
| 115 | M21   | 2     | 1         | 0.079 | 0.072 | 0.375  | 0.483      | 0.955 |
| 115 | M21   | 2     | 2         | 1.093 | 0.703 | 0.625  | 0.483      | 0.955 |
| 115 | M22   | 1     | 1         | 0.057 | 0.050 | 0.202  | 0.835      | 0.900 |
| 115 | M22   | 1     | 2         | 0.461 | 0.238 | 0.798  | 0.835      | 0.900 |
| 115 | M22   | 2     | 1         | 1.281 | 0.221 | 0.696  | 0.165      | 0.465 |
| 115 | M22   | 2     | 2         | 2.341 | 0.223 | 0.304  | 0.165      | 0.465 |
| 163 | M21   | 1     | 1         | 0.062 | 0.050 | 0.630  | 0.707      | 0.787 |
| 163 | M21   | 1     | 2         | 0.258 | 0.110 | 0.370  | 0.707      | 0.787 |
| 163 | M21   | 2     | 1         | 1.486 | 0.784 | 1.000  | 0.293      | 0.469 |
| 163 | M22   | 1     | 1         | 0.064 | 0.050 | 0.767  | 0.619      | 0.763 |
| 163 | M22   | 1     | 2         | 0.238 | 0.050 | 0.233  | 0.619      | 0.763 |
| 163 | M22   | 2     | 1         | 0.460 | 0.209 | 0.446  | 0.381      | 0.603 |
| 163 | M22   | 2     | 2         | 1.834 | 0.631 | 0.554  | 0.381      | 0.603 |
| 287 | M21   | 1     | 1         | 0.118 | 0.068 | 0.639  | 0.804      | 0.890 |
| 287 | M21   | 1     | 2         | 0.432 | 0.173 | 0.361  | 0.804      | 0.890 |
| 287 | M21   | 2     | 1         | 1.753 | 0.649 | 1.000  | 0.196      | 0.552 |
| 287 | M22   | 1     | 1         | 0.164 | 0.086 | 0.563  | 0.442      | 1.000 |
| 287 | M22   | 1     | 2         | 0.466 | 0.135 | 0.437  | 0.442      | 1.000 |
| 287 | M22   | 2     | 1         | 0.105 | 0.061 | 0.582  | 0.558      | 0.989 |
| 287 | M22   | 2     | 2         | 1.563 | 0.741 | 0.418  | 0.558      | 0.989 |
| 363 | M21   | 1     | 1         | 0.459 | 0.238 | 1.000  | 0.317      | 0.857 |
| 363 | M21   | 2     | 1         | 0.139 | 0.092 | 0.705  | 0.683      | 0.940 |
| 363 | M21   | 2     | 2         | 1.377 | 0.613 | 0.295  | 0.683      | 0.940 |
| 363 | M22   | 1     | 1         | 0.345 | 0.171 | 0.753  | 0.495      | 0.892 |
| 363 | M22   | 1     | 2         | 0.927 | 0.347 | 0.247  | 0.495      | 0.892 |
| 363 | M22   | 2     | 1         | 0.119 | 0.082 | 0.764  | 0.505      | 0.902 |
| 363 | M22   | 2     | 2         | 1.692 | 0.510 | 0.236  | 0.505      | 0.902 |

In a mixture-emission HMM, a new component is drawn conditional on the aggregate state at each observation. Components therefore share the aggregate transition probabilities and do not carry a separate persistence parameter. For elk 163, 100% of G3 state 1 and 87.2% of G3 state 2 are assigned to M21 state 1 by the respective Viterbi paths; G3 state 3 maps entirely to M21 state 2. The assignment is descriptive and does not establish which partition is biologically correct. The retained M22–G3 comparison for all animals is shown in Figure S6.

For elk 163, M21–G3 BIC differences remain negative at all three bounds. After allowing for state-label permutation, the M21 Viterbi paths at bounds 0.025 and 0.10 agree with the primary

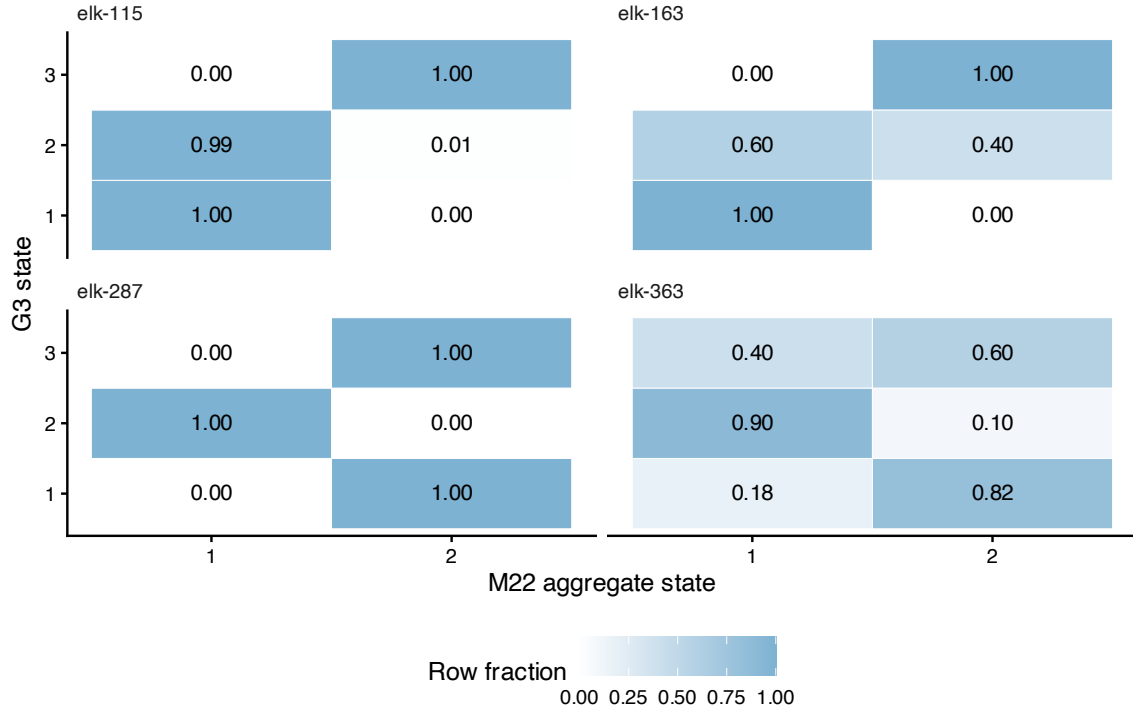


Figure S6: Row proportions mapping full-series G3 Viterbi states to M22 states for each animal, at standard-deviation bound 0.05, using the refined fits.

Table S9: BIC differences across standard-deviation bounds. Negative entries favor the mixture model. The first and third contrasts hold the total number of Gaussian components fixed.

| Elk | Bound | M21 – G3 | M22 – G3 | M22 – G4 |
|-----|-------|----------|----------|----------|
| 115 | 0.025 | 11.5     | 13.8     | 4.8      |
| 163 | 0.025 | -11.4    | -12.7    | -28.4    |
| 287 | 0.025 | 20.5     | -8.7     | -39.2    |
| 363 | 0.025 | -8.4     | -8.0     | -9.9     |
| 115 | 0.050 | -2.0     | 2.8      | -2.7     |
| 163 | 0.050 | -7.8     | -10.8    | -30.2    |
| 287 | 0.050 | 20.5     | -8.7     | -39.2    |
| 363 | 0.050 | -8.4     | -8.0     | -24.6    |
| 115 | 0.100 | -5.4     | 1.4      | -16.3    |
| 163 | 0.100 | -7.2     | -5.1     | -33.1    |
| 287 | 0.100 | 11.4     | -9.9     | -38.4    |
| 363 | 0.100 | -9.9     | -6.3     | -35.3    |

path on 98.1% and 93.0% of intervals. Thus the broad allocation of low and high displacements is relatively stable, even though the lower-state density loses a mode at the largest bound. Mode counts are computed on a 50,001-point grid spanning all fitted component means plus or minus five standard deviations. Counts describe the fitted real-line density; they are not estimates of the number of behaviours.

For the four-peak density of elk 115, the component means at bound 0.05 are 0.057, 0.461, 1.281 and 2.341, with standard deviations 0.050, 0.238, 0.221 and 0.223. The density is weighted by posterior state occupation, rather than assumed stationary occupation. All four peaks can therefore be traced to specified emission components. Figure S7 shows the sensitivity of this shape to the fitting bound. The selected M22 density has three modes at 0.025 and four at 0.05 and 0.10. The first component is at the bound in the latter two fits. Neither a four-mode interpretation nor a claim that only three modes are possible is supported.

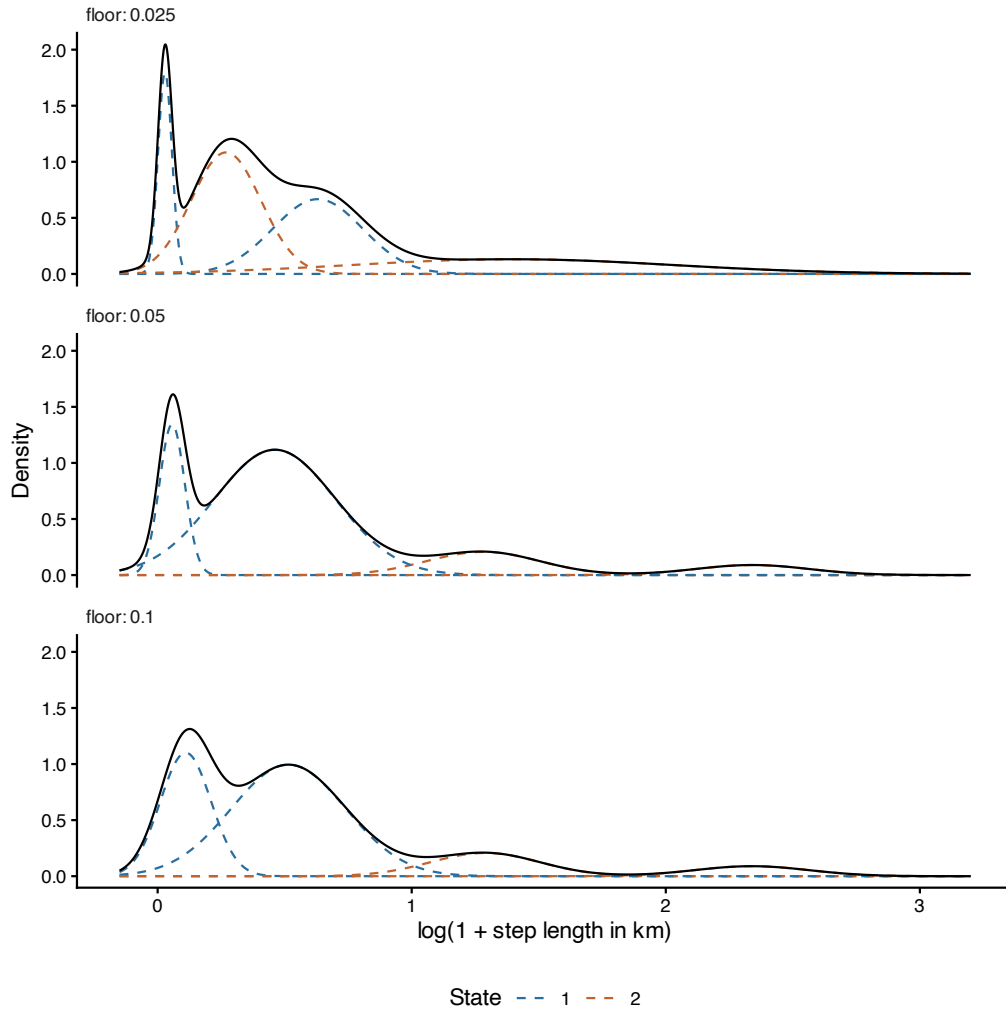


Figure S7: Elk 115: M22 density decomposition at three standard-deviation bounds. Dashed curves are the four weighted component contributions, colours identify aggregate states, and black curves give the total density.

## S5 Elk movement: joint models and alternative emissions

### S5.1 Measurements, observation likelihood and model counts

The original text archive contains the 735 locations used in the Gaussian benchmark. Removing its 370 blank rows and matching animal IDs and all coordinate pairs verifies correspondence with the package data. The new analysis uses published daily movement rates, which adjust displacement for the observation interval, rather than reconstructing unavailable timestamps. Rate and angle refer to the outgoing displacement at a location. The first angle is unavailable; angles involving the zero-length displacement of elk 363, or its subsequent displacement, are omitted. There are 731 intervals, one missing rate, 14 reported zero rates and six missing angles after these omissions. The four animals remain separate likelihoods.

Rates are published to two decimal places. We treat each value as rounded to the nearest 0.01 km/day. With rate distribution function  $F$ , the contribution of value  $r$  is  $F(r + 0.005) - F(\max(0, r - 0.005))$ . This observation model avoids assigning an artificial positive rate to a zero. It is a rounding assumption, not a model for positional error. Missing response components contribute a factor of one. The transition recursion continues across their intervals. Turning angles use a wrapped-Cauchy density with estimated mean direction and concentration. We condition on an initial state distribution estimated separately for each trajectory.

The four rate families are gamma (mean and shape), Weibull (scale and shape), lognormal (log mean and log standard deviation), and a normal distribution on  $\log(1 + r)$  truncated below zero. All have two emission parameters per component. The joint families add two angular parameters per component. M21 and M22 have component counts (2, 1) and (2, 2). H2, H3 and H4 have one component per state. Their rate-only parameter counts are 7, 14, 23, 10 and 13 for H2, H3, H4, M21 and M22; the corresponding joint counts are 11, 20, 31, 16 and 21. BIC uses the number of intervals with at least one observed response component. It is used descriptively, particularly for mixtures and near-boundary transition estimates.

The positive-family alternatives change the assessment of elk 163. In rate-only fits, Weibull H2 has BIC 1680.3, compared with 1684.4 for M21. In the joint Weibull analysis H2 has BIC 2254.3, below M21 at 2264.8. The Gaussian mixture result for that animal is therefore sensitive to emission family. For elk 287, M21 has lower joint BIC than H3 under all four rate families; this is evidence for a compact fitted representation, not for identification of the true latent states.

### S5.2 Numerical optimization and independent checks

The binned likelihood is evaluated with log distribution functions and stable differences in distribution tails. A scaled forward recursion combines these probabilities with the optional angular densities. Independent checks compare interval probabilities with R distribution functions, the forward likelihood with explicit enumeration of short state sequences, and the two-state mixture with its exact three-state expansion. Forecast scores telescope to likelihood differences, and changing a future observation does not change earlier scores. Posterior decoding is checked against an independently

Table S10: Full-series BIC for rate-only models. Every comparison uses the same rounding intervals and missing-data treatment.

| Elk | Family        | H2     | H3     | H4     | M21    | M22    |
|-----|---------------|--------|--------|--------|--------|--------|
| 115 | Gamma         | 2190.9 | 2159.8 | 2192.4 | 2167.2 | 2178.3 |
| 163 | Gamma         | 1682.0 | 1691.0 | 1723.2 | 1683.9 | 1695.0 |
| 287 | Gamma         | 1742.5 | 1726.0 | 1759.8 | 1715.1 | 1725.5 |
| 363 | Gamma         | 2350.9 | 2334.3 | 2344.5 | 2318.9 | 2333.9 |
| 115 | Weibull       | 2180.1 | 2157.0 | 2191.0 | 2168.6 | 2178.1 |
| 163 | Weibull       | 1680.3 | 1691.1 | 1722.7 | 1684.4 | 1694.4 |
| 287 | Weibull       | 1744.5 | 1722.7 | 1759.0 | 1712.3 | 1723.8 |
| 363 | Weibull       | 2351.2 | 2335.1 | 2345.4 | 2319.6 | 2334.6 |
| 115 | Lognormal     | 2161.4 | 2173.7 | 2197.1 | 2170.5 | 2182.3 |
| 163 | Lognormal     | 1684.1 | 1696.7 | 1729.7 | 1689.6 | 1700.2 |
| 287 | Lognormal     | 1747.3 | 1744.1 | 1769.4 | 1732.1 | 1737.3 |
| 363 | Lognormal     | 2316.8 | 2331.5 | 2349.0 | 2322.7 | 2336.4 |
| 115 | Trunc. normal | 2163.6 | 2161.6 | 2191.6 | 2169.9 | 2181.1 |
| 163 | Trunc. normal | 1689.6 | 1696.3 | 1726.6 | 1688.6 | 1700.1 |
| 287 | Trunc. normal | 1749.4 | 1725.1 | 1760.0 | 1714.7 | 1724.9 |
| 363 | Trunc. normal | 2344.7 | 2339.0 | 2347.0 | 2333.6 | 2339.0 |

Table S11: Full-series BIC for joint rate and angle models. Every comparison uses the same rounding intervals and missing-data treatment.

| Elk | Family        | H2     | H3     | H4     | M21    | M22    |
|-----|---------------|--------|--------|--------|--------|--------|
| 115 | Gamma         | 2891.8 | 2862.2 | 2883.2 | 2865.5 | 2871.9 |
| 163 | Gamma         | 2256.2 | 2275.6 | 2309.5 | 2264.3 | 2279.8 |
| 287 | Gamma         | 2319.8 | 2307.8 | 2334.4 | 2296.6 | 2308.8 |
| 363 | Gamma         | 3129.4 | 3120.4 | 3129.4 | 3107.6 | 3127.2 |
| 115 | Weibull       | 2881.0 | 2861.0 | 2883.9 | 2866.5 | 2873.3 |
| 163 | Weibull       | 2254.3 | 2276.1 | 2314.5 | 2264.8 | 2277.9 |
| 287 | Weibull       | 2319.8 | 2304.2 | 2332.8 | 2293.2 | 2307.1 |
| 363 | Weibull       | 3129.5 | 3122.2 | 3128.9 | 3109.9 | 3130.0 |
| 115 | Lognormal     | 2862.4 | 2870.1 | 2885.7 | 2859.6 | 2873.4 |
| 163 | Lognormal     | 2258.6 | 2281.3 | 2316.3 | 2273.7 | 2278.8 |
| 287 | Lognormal     | 2326.7 | 2322.1 | 2348.4 | 2312.4 | 2320.4 |
| 363 | Lognormal     | 3114.7 | 3116.4 | 3141.7 | 3118.1 | 3123.6 |
| 115 | Trunc. normal | 2866.1 | 2862.5 | 2886.4 | 2866.0 | 2875.5 |
| 163 | Trunc. normal | 2264.6 | 2281.2 | 2316.3 | 2271.1 | 2288.8 |
| 287 | Trunc. normal | 2322.2 | 2306.4 | 2335.6 | 2295.3 | 2308.3 |
| 363 | Trunc. normal | 3127.3 | 3126.4 | 3134.8 | 3117.2 | 3130.2 |

implemented forward–backward recursion.

Full-series fits use 40 random starts, screened for 70 iterations and 180 function evaluations. At least eight are refined for up to 1200 iterations and 3500 evaluations at relative tolerance  $10^{-9}$ , followed by a 2000-iteration restart at  $10^{-10}$ . Selected Weibull and gamma joint fits use 40 further random starts plus estimates from sensitivity fits, all re-evaluated on the original full series. The largest improvement from these additional starts was 8.26 log-likelihood units for gamma M21 on elk 287. This illustrates why an optimizer’s stopping code alone does not establish that competing local solutions were adequately explored. The final fits, all start-level results and these improvements are retained.

Training-prefix fits initially use 20 starts. The primary Weibull and gamma joint comparisons receive 80 additional starts each, using only that prefix’s earlier estimates and the nested M21 expansion for H3. Some training fits also improve materially. No full-series or later-prefix parameter is supplied as a training start. The full grid contains 160 full-series and 480 training-prefix fits. An optimization audit re-evaluates every likelihood and computes numerical projected gradients at the bounds. Optimizer messages and gradient magnitudes are recorded separately; a successful stopping code is not treated as proof of a global maximum. Of the 640 selected fits, 639 have a zero optimizer code. One gamma H2 training fit for elk 363 retains a false-convergence message after restarts; its maximum projected gradient is  $3.1 \times 10^{-5}$ . The largest projected gradient over the grid is 0.00472. The warning remains in the saved results.

Logit parameters are bounded by  $\pm 12$ . The emission ranges are deliberately broad: positive-family location parameters on a log scale lie in  $[-9, 5]$ ; gamma shape in  $[0.05, 200]$ ; Weibull shape in  $[0.1, 30]$ ; lognormal and truncated-normal standard deviations in  $[0.05, 4]$ . The truncated-normal location lies in  $[-3, 5]$ . Wrapped-Cauchy concentration lies in  $[0, 0.95]$ , with 0.90 and 0.98 as sensitivity bounds. Numerical boundary estimates are reported, not interpreted as interior parameter estimates.

### S5.3 Five temporal blocks and predictive calibration

The cutoffs are  $\lfloor fT \rfloor$  for  $f = 0.5, 0.6, 0.7, 0.8, 0.9$ . Each fit scores the following tenth of the record, with the final block ending at  $T$ . Thus the 97, 79, 82 and 109 scored intervals do not overlap. Training sets are nested, so block comparisons are dependent. Model families and the worked case were examined retrospectively; these are not untouched confirmation sets. Results for every family and animal are retained below.

Rate-only and joint log scores concern different response spaces and must not be compared directly. Rate predictions conditional on past rates and angles are also saved separately. For binned rates, randomized PIT values are drawn uniformly between the predictive distribution function at the lower and upper rounding bounds. One saved randomization is shared across models for each animal and interval. The central 90% PIT fraction is a descriptive calibration check, not an independence test or an interval estimate for the model’s coverage probability. For elk 287 under Weibull emissions, the within-block lag-one correlations of normal-score PIT residuals are 0.037

Table S12: Joint predictive log scores over five disjoint blocks. Parameters are estimated on the preceding training prefix only. These retrospective comparisons are exploratory.

| Elk | Family        | H2     | H3     | M21    | M21-H3 |
|-----|---------------|--------|--------|--------|--------|
| 115 | gamma         | -721.4 | -732.5 | -734.2 | -1.7   |
| 115 | lognormal     | -740.9 | -733.2 | -727.2 | 6.0    |
| 115 | Trunc. normal | -736.5 | -726.7 | -728.6 | -1.9   |
| 115 | weibull       | -721.4 | -735.0 | -738.0 | -3.0   |
| 163 | gamma         | -520.4 | -524.6 | -525.9 | -1.3   |
| 163 | lognormal     | -514.2 | -514.6 | -523.0 | -8.3   |
| 163 | Trunc. normal | -517.1 | -525.4 | -526.2 | -0.8   |
| 163 | weibull       | -521.1 | -528.2 | -526.6 | 1.6    |
| 287 | gamma         | -560.6 | -560.4 | -561.1 | -0.7   |
| 287 | lognormal     | -558.0 | -554.9 | -553.9 | 1.0    |
| 287 | Trunc. normal | -561.9 | -561.9 | -564.6 | -2.7   |
| 287 | weibull       | -560.9 | -561.0 | -562.2 | -1.3   |
| 363 | gamma         | -792.8 | -802.1 | -794.5 | 7.6    |
| 363 | lognormal     | -804.7 | -789.8 | -789.8 | 0.0    |
| 363 | Trunc. normal | -784.9 | -788.8 | -788.2 | 0.6    |
| 363 | weibull       | -791.7 | -796.1 | -785.3 | 10.8   |

(M21), 0.009 (H3) and 0.062 (H2). These are descriptive dependence checks without a significance claim. The central 90% PIT fraction is 98.8% for M21 and H2 and 97.6% for H3, indicating conservative rate predictions in the scored part of this record.

The Gaussian benchmark was also extended to the same five cutoffs with its original 200-start protocol. M21 minus G3 score differences are 1.12, 2.04,  $-16.43$  and  $-6.22$  for elk 115, 163, 287 and 363, respectively. The corresponding per-block and per-observation results are retained in `empirical/application_review/gaussian_validation_blocks.csv` and `empirical/application_review/gaussian_validation_predictions.csv`. Extending the cutoffs does not turn the Gaussian comparison into a uniform predictive advantage for a mixture.

#### S5.4 Phase interpretation, resolution and angular heterogeneity

The joint Weibull M21 fit for elk 287 assigns intervals 1–91 to a mobile phase and 92–163 to a localized phase. Its two mobile components combine long directed movements with short, more diffusely oriented movements. The localized phase’s mean rate lies between the means of those components. It therefore does not represent simply the shortest movements. Maximum pairwise distances among the locations spanned by each decoded phase are 89.77 and 1.56 km. Median distance to water is 0.460 and 2.071 km. These are descriptions of the same record, not withheld behavioural measurements. Raw habitat codes are not interpreted: their correspondence with the original article’s recoded categories is not documented in the data table.

To check sensitivity to the smallest movements, values below 0.05 km/day are grouped into  $[0, 0.045)$ , the union of their original rounding bins. The analogous grouping below 0.10 is  $[0, 0.095)$ . Angles involving either small displacement are omitted at each resolution. This changes the observation being scored, so BIC values are comparable only within the same coarsening scenario.

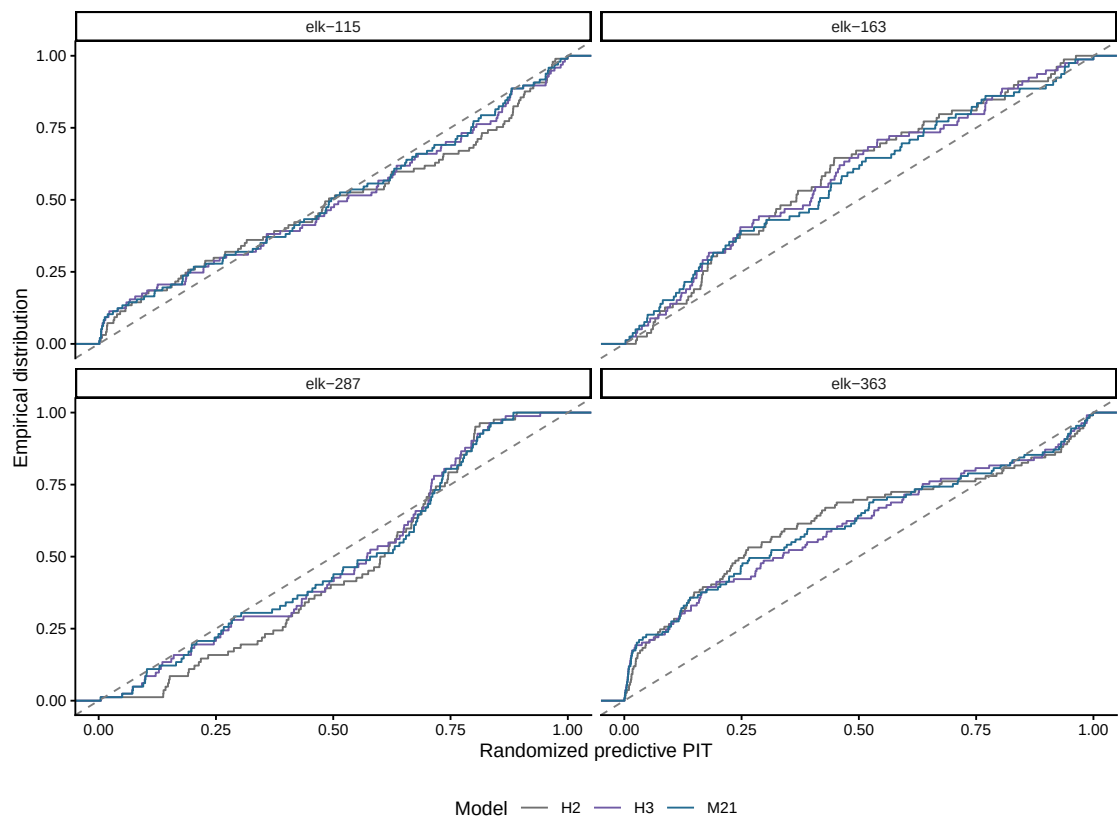


Figure S8: Joint Weibull-model predictive PIT distributions, using the five scored blocks and a common randomization for rounded rates. The diagonal is the uniform reference. Deviations remain despite favorable full-data BIC in some comparisons.

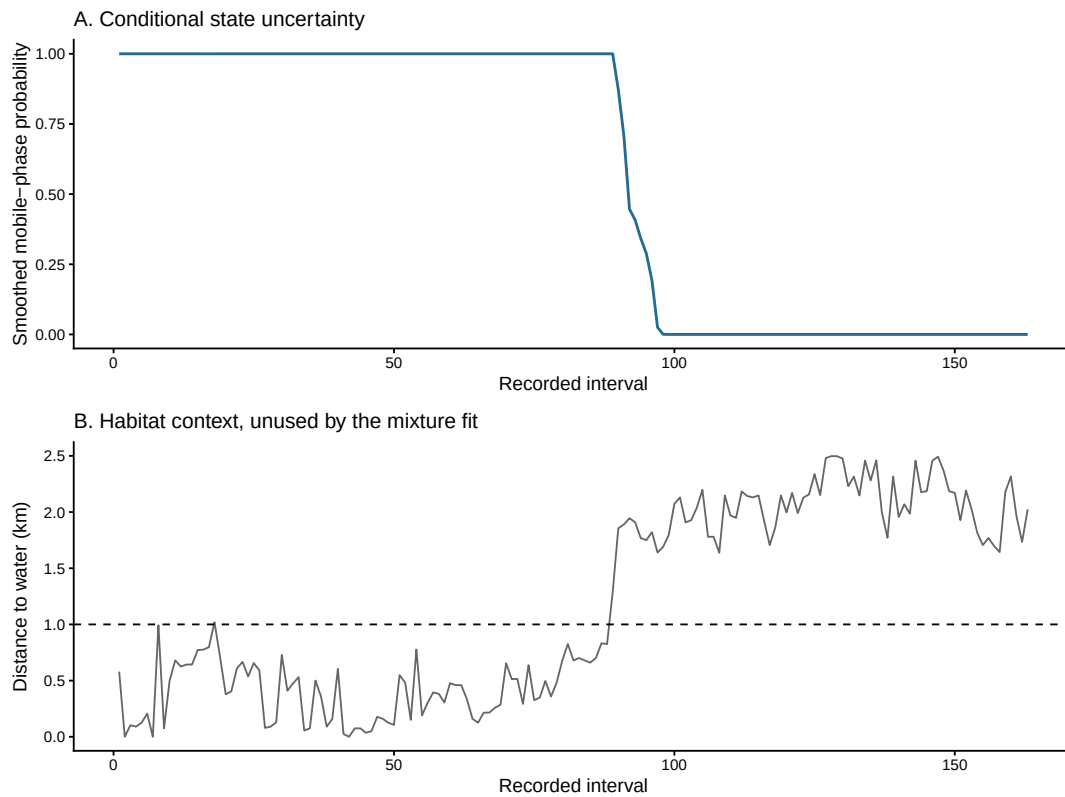


Figure S9: Elk 287. Panel A gives the mobile-phase posterior conditional on the fitted M21 parameters; it does not include parameter uncertainty. Panel B shows distance to water, which was not fitted by this mixture model. The dashed line is 1 km and is a descriptive reference, not a classification threshold.

Table S13: Elk 287 sensitivity: BIC comparisons within each row and M21 path agreement with the corresponding uncoarsened family. Coarse 005 and 010 group rates below 0.05 and 0.10 km/day and omit angles involving those small movements. Scores across coarsening scenarios concern different observations and must not be compared.

| Family  | Check        | M21    | H2     | H3     | Agreement (%) |
|---------|--------------|--------|--------|--------|---------------|
| weibull | coarse 005   | 2183.5 | 2210.8 | 2194.6 | 96.9          |
| gamma   | coarse 005   | 2185.3 | 2210.6 | 2196.6 | 96.9          |
| weibull | coarse 010   | 1912.4 | 1930.3 | 1922.9 | 96.9          |
| gamma   | coarse 010   | 1914.6 | 1927.1 | 1925.0 | 96.9          |
| weibull | shared angle | 2302.3 | 2319.8 | 2304.2 | 100.0         |
| gamma   | shared angle | 2305.1 | 2319.8 | 2307.8 | 98.8          |
| weibull | rho 090      | 2293.2 | 2319.8 | 2304.2 | 100.0         |
| gamma   | rho 090      | 2296.6 | 2319.8 | 2307.8 | 100.0         |
| weibull | rho 098      | 2293.2 | 2319.8 | 2304.2 | 100.0         |
| gamma   | rho 098      | 2296.6 | 2319.8 | 2307.8 | 100.0         |

It does not reproduce a full location-error likelihood. Both coarsened Weibull M21 fits retain two periods, with 96.9% agreement with the original decoding. All other animals are also refitted; their complete results remain in the reproducibility files.

Sharing one angular distribution across both mobile components gives Weibull M21 14 parameters and BIC 2302.3, compared with 2293.2 for separate component directions. This comparison suggests that the mixture’s role includes a rate–direction association within the mobile phase. It is not merely an additional marginal mode. The concentration-bound checks leave the primary Weibull phase path unchanged.

Only one phase change is observed. The M21 return probability is at the logit boundary, approximately  $6.1 \times 10^{-6}$ . Bounds 8, 12 and 16 give return probabilities approximately  $3.4 \times 10^{-4}$ ,  $6.1 \times 10^{-6}$  and  $1.1 \times 10^{-7}$ . Their log-likelihoods differ by about 0.02 and their Viterbi paths are identical. The stable decoding does not identify an eventual return time or a recurrent ecological switching process. The near-absorbing interpretation is kept explicit.

## S5.5 A competing covariate explanation

A two-state joint model uses separate logistic regressions for the two transition rows, with an intercept and slope on the standardized  $\log(1 + \text{distance to water in km})$ . Standardization uses the training prefix only. The covariate at the departure location of the current interval determines its transition probabilities. For validation, these are conditional forecasts given the current observed location, not unconditional forecasts of the next location. Zero slopes reproduce the homogeneous H2 likelihood exactly, which is checked numerically; a short time-varying example is also checked by enumerating every state sequence.

This model has 13 parameters. For elk 287 the Weibull BIC is 2312.9, an improvement over homogeneous H2 (2319.8), but above M21 (2293.2). Gamma gives a similar covariate-model BIC, 2312.8. All four animals and five training cutoffs are evaluated in `empirical/application_revi`

ew/water\_comparison.csv. These separate fits differ from the pooled multi-animal analysis of Patterson et al. (2017), <https://doi.org/10.1007/s10182-017-0302-7>, which motivated this alternative. One distance covariate cannot exclude other covariates, explicit phase-change models, time-varying dynamics or measurement error. No causal water effect, habitat-selection estimate or new species-level behaviour is inferred.

## S6 Reproduction sequence

For the genomic application, run from the reproducibility-package root:

```
Rscript genomics/scripts/test_genomic_results.R
Rscript genomics/scripts/test_symmetric_genomic_hmm.R
Rscript genomics/scripts/test_genomic_g4.R
Rscript genomics/scripts/evaluate_genomic_pilot.R
Rscript genomics/scripts/genomic_diagnostics.R
Rscript numerics/scripts/render_publication_tables.R
```

The supplied processed inputs allow this numerical verification without reprocessing the microarrays. The genomic README documents a complete seeded refit and the more demanding route from the four original CEL files, including exact annotation versions and the R package inventory. It also records the separate development, validation and confirmation sequence. Saved fits are evidence for the reported calculations, not substitutes for these reproduction instructions.

For the retained elk analyses, run the following commands from the reproducibility-package root, using R and a C++ compiler supported by Rcpp:

```
Rscript -e 'renv::restore()'
Rscript numerics/scripts/run_revision_validation.R
Rscript numerics/scripts/refine_revision_sensitivity.R
Rscript numerics/scripts/run_revision_validation.R
Rscript numerics/scripts/check_elk_context.R
Rscript numerics/scripts/render_revision_outputs.R
Rscript numerics/scripts/test_revision.R
```

The repeated validation command reads the cached fitted objects and refreshes the summary tables after the cross-bound refinements; it does not refit cached models. Remove the generated `empirical/revision2/fits` directory in a separate working copy to rerun all estimates from their seeds. The corrected simulation uses `numerics/scripts/run_optimization_audit.R` and `numerics/scripts/refine_reference_simulation.R`, with its protocol specified in Section S2. The refinement bootstrap is reproduced with `genomics/scripts/genomic_refinement_bootstrap.R`. These scripts retain checkpointed fits; the package README explains how to rerun them from seeds.

The Gaussian analyses use the existing R EM updates with a compiled version of the scaled forward-backward recursion. The compiled and R recursions are checked against one another, and the tests also cover exact refinement, held-out filtering, likelihood monotonicity, variance-bound sensitivity, normalization, parameter counts and correspondence between saved fits and numerical tables. No raw coordinate table is redistributed. Numerical inputs, derived outputs, model objects, source hashes and session information are documented in `empirical/revision2`.

For the expanded movement analysis, run the following additional sequence. Cached fits are reused; use a separate copy without the generated `empirical/application_review` directory for a complete seeded rerun. The input file is downloaded from the archived public source and its SHA-256 digest is checked before use.

```
Rscript numerics/scripts/test_movement_review.R
Rscript numerics/scripts/run_movement_review.R
Rscript numerics/scripts/run_movement_review.R joint
Rscript numerics/scripts/run_movement_validation.R
Rscript numerics/scripts/run_movement_validation.R joint
Rscript numerics/scripts/refine_movement_review.R
Rscript numerics/scripts/run_movement_sensitivity.R
Rscript numerics/scripts/refine_movement_starts.R
Rscript numerics/scripts/refine_movement_validation.R
Rscript numerics/scripts/refine_movement_review.R
Rscript numerics/scripts/run_movement_review.R
Rscript numerics/scripts/run_movement_review.R joint
Rscript numerics/scripts/run_movement_validation.R
Rscript numerics/scripts/run_movement_validation.R joint
Rscript numerics/scripts/run_movement_sensitivity.R
Rscript numerics/scripts/run_water_comparison.R
Rscript numerics/scripts/run_extended_gaussian_validation.R
Rscript numerics/scripts/describe_movement_review.R
Rscript numerics/scripts/check_movement_phase.R
Rscript numerics/scripts/test_movement_results.R
Rscript numerics/scripts/render_movement_review.R
```